

Hunters Hill High School
Extension 2, Mathematics

Trial Examination, 2015



Hunters Hill
High School

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- The marks for each question are shown on the paper
- Show all necessary working in questions 11-16

Total Marks: 100

Section I Page 3 – 7

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

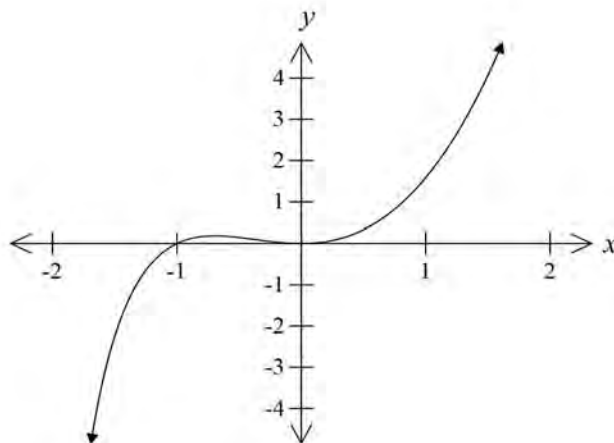
Section II Pages 8 – 14

90 marks

- Attempt Questions 11-16
- Allow about 2 hour 45 minutes for this section

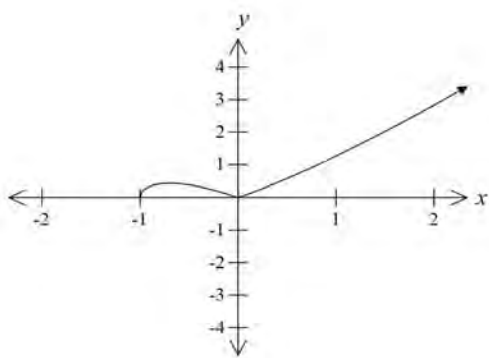
Section I MULTIPLE CHOICE: Write the correct alternative on your writing paper.

- 1 The diagram shows the graph of the function $y = f(x)$.

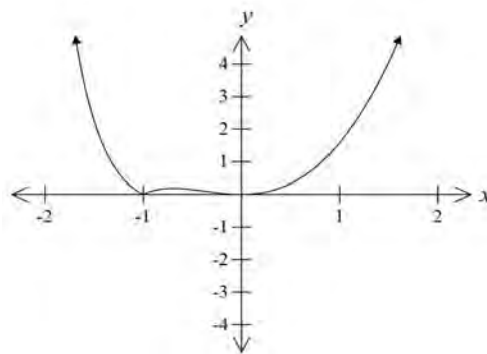


Which of the following is the graph of $y = f(x)^2$?

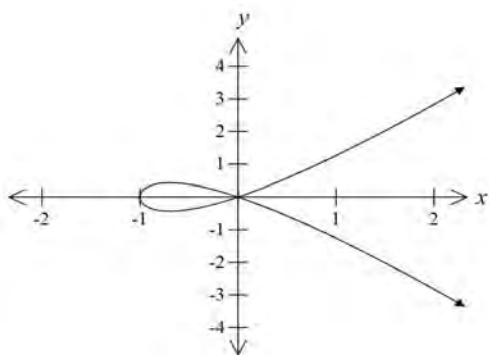
(A)



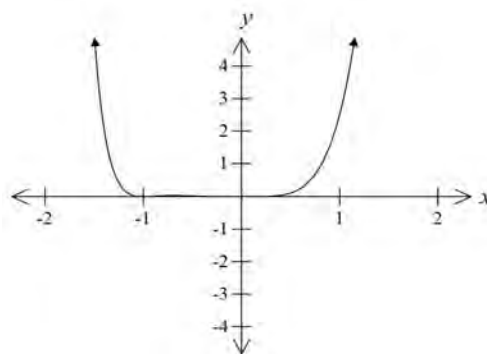
(B)



(C)



(D)

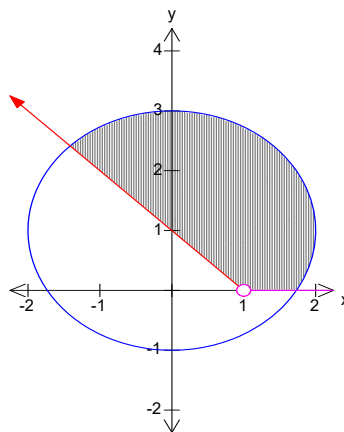


- 2 Consider the hyperbola with the equation $\frac{x^2}{144} - \frac{y^2}{25} = 1$.

What are the equations of the directrices?

- (A) $x = \pm \frac{13}{144}$
 (B) $x = \pm \frac{13}{25}$
 (C) $x = \pm \frac{25}{13}$
 (D) $x = \pm \frac{144}{13}$

- 3 Consider the Argand diagram below.



Which inequality could define the shaded area?

- (A) $|z - i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$ (B) $|z + i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$
 (C) $|z - i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{\pi}{4}$ (D) $|z + i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{\pi}{4}$

4 Which of the following is an expression for $\int \frac{2}{x^2 + 4x + 13} dx$?

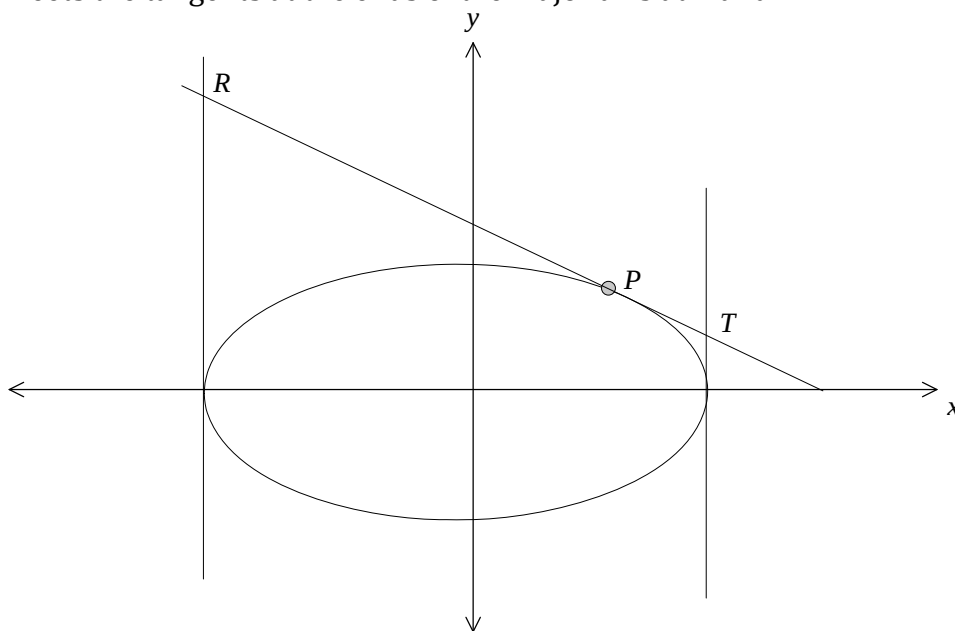
(A) $\frac{1}{3} \tan^{-1} \frac{(x+2)}{3} + c$

(B) $\frac{2}{3} \tan^{-1} \frac{(x+2)}{3} + c$

(C) $\frac{1}{9} \tan^{-1} \frac{(x+2)}{9} + c$

(D) $\frac{2}{9} \tan^{-1} \frac{(x+2)}{9} + c$

5 The point P lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b > 0$. The tangent at P meets the tangents at the ends of the major axis at R and T .



What is the equation of the tangent at P ?

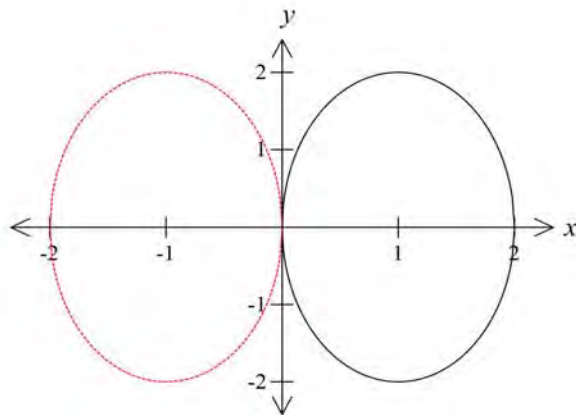
(A) $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$

(B) $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$

(C) $\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$

(D) $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$

- 6 The region enclosed by the ellipse $(x-1)^2 + \frac{y^2}{4} = 1$ is rotated about the y axis to form a solid.



What is the correct expression for volume of this solid using the method of slicing?

- (A) $V = \int_{-2}^2 \pi \sqrt{1-y^2} dy$
- (B) $V = \int_{-2}^2 2\pi \sqrt{1-y^2} dy$
- (C) $V = \int_{-2}^2 \pi \sqrt{4-y^2} dy$
- (D) $V = \int_{-2}^2 2\pi \sqrt{4-y^2} dy$
- 7 The polynomial equation $x^3 - 3x^2 - x + 2 = 0$ has roots α , β and γ . Which of the following polynomial equations have roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$?
- (A) $x^3 - x^2 - 3x + 1 = 0$
- (B) $x^3 - 2x^2 - 3x + 1 = 0$
- (C) $2x^3 - x^2 - 3x + 1 = 0$
- (D) $2x^3 - 2x^2 - 3x + 1 = 0$

- 8 What is the derivative of $\sin^{-1} x - \sqrt{1-x^2}$?
- (A) $\frac{\sqrt{1+x}}{\sqrt{1-x}}$
- (B) $\frac{\sqrt{1+x}}{1-x}$
- (C) $\frac{1+x}{\sqrt{1-x}}$
- (D) $\frac{1+x}{1-x}$
- 9 What are the values of real numbers p and q such that $1-i$ is a root of the equation $z^3 + pz + q = 0$?
- (A) $p = -2$ and $q = -4$
- (B) $p = -2$ and $q = 4$
- (C) $p = 2$ and $q = 4$
- (D) $p = 2$ and $q = 4$
- 10 A particle of mass m is moving in a straight line under the action of a force.
- $$F = \frac{m}{x^3}(6-10x)$$
- What of the following is an expression for its velocity in any position, if the particle starts from rest at $x = 1$?
- (A) $v = \pm \frac{1}{x} \sqrt{(-3+10x-7x^2)}$
- (B) $v = \pm x \sqrt{(-3+10x-7x^2)}$
- (C) $v = \pm \frac{1}{x} \sqrt{2(-3+10x-7x^2)}$
- (D) $v = \pm x \sqrt{2(-3+10x-7x^2)}$

Total Marks – 90**Section II****Attempt Questions 11-16****All Questions are of equal value****Marks****Question 11 (15 marks)** Begin a NEW sheet of paper.

- (a) By using the method of partial fractions, show that

4

$$\int \frac{dx}{x^2 - 1} = \ln \sqrt{\frac{x-1}{x+1}} + c$$

- (b) Use the substitution $u = \cos x$ to evaluate

4

$$\int_0^1 \sqrt{1-x^2} \, dx$$

- (c) If $I = \int e^x \sin x \, dx$

Find I using the method of integration by parts.

4

- (d) Evaluate

$$\int_0^{\frac{\pi}{2}} \cos x \sin^3 x \, dx$$

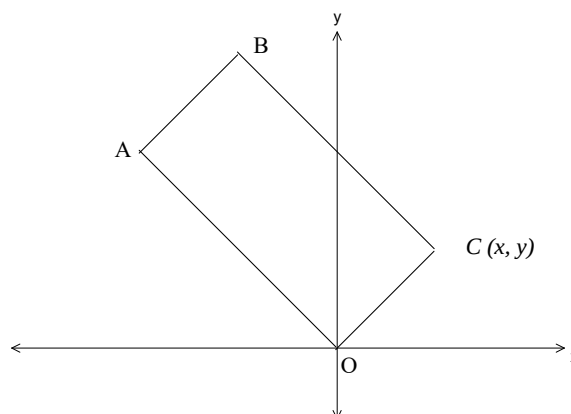
3**End of Question 11**

Question 12 (15 marks) Begin a NEW sheet of paper.**Marks**(a) Given $A = 3 - 4i$ and $B = \sqrt{3} + i$.(i) Find AB in $x + iy$ form 1(ii) Find $\frac{A}{B}$ in $x + iy$ form 1(iii) Find $\sqrt{A}$ in $x + iy$ form 3(iv) Find B in modulus- argument form 2(v) Hence find B^4 in $x + iy$ form 1

(b) On separate Argand diagrams sketch the following loc

(i) $2 \geq |z| \geq 1$ 1(ii) $\frac{3\pi}{4} > \arg z > \frac{\pi}{4}$ 1(iii) $3 \geq \operatorname{Re} Z \geq 0$ and $3 \geq \operatorname{Im} Z \geq 1$ 2(c) On the Argand diagram shown OABC is a rectangle with the length OA being twice OC. OC represents the complex number $x + iy$.

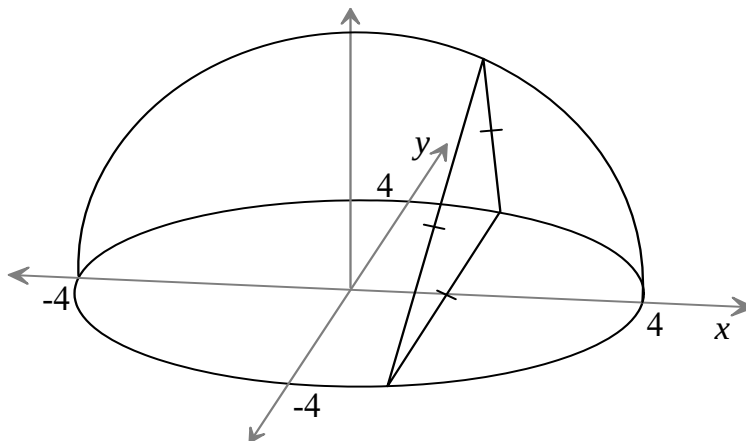
Find the complex number represented by

(i) OA 1(ii) OB 1(iii) BC 1**End of Question 12**

Question 13 (15 marks) Begin a NEW sheet of paper.**Marks**

- (a) For the curve with equation $x^2 + 3xy - y^2 = 13$, determine the gradient of the tangent at the point $(2, 3)$ on the curve. 4

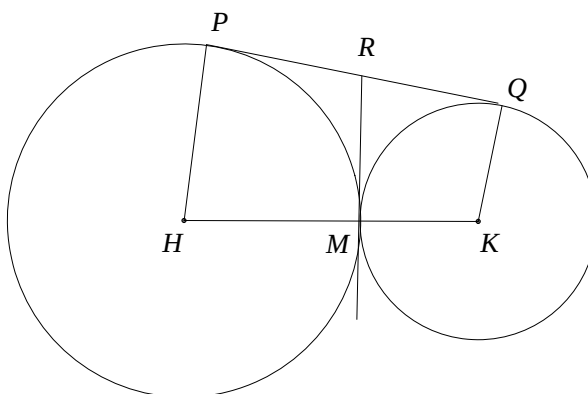
(b)



The diagram above shows a solid which has the circle $x^2 + y^2 = 16$ as its base. The cross-section perpendicular to the x axis is an equilateral triangle.

- (i) Show that the cross-sectional area of the equilateral triangle is given by $A(x) = \sqrt{3}(16 - x^2)$ 3
- (ii) Calculate the volume of the solid. 3
- (c) Shown are two circles centres H and K which touch at M . PQ and RM are common tangents.

COPY
DIAGRAM



- (i) Show that quadrilaterals $HPRM$ and $MRQK$ are cyclic. 2
- (ii) Prove that triangles PRM and MKQ are similar. 3

End of Question 13

Question 14 (15 marks) Begin a NEW sheet of paper.		Marks
(a)	The equation $P(x) = x^3 + 3x^2 - 24x + k = 0$ has a double root. Find the possible values of k .	2
(b)	The roots of $x^3 + 3px + q = 0$ are α, β and γ , (none of which are equal to 0).	
(i)	Find the monic equation with roots $\frac{\beta\gamma}{\alpha}, \frac{\alpha\gamma}{\beta}$ and $\frac{\alpha\beta}{\gamma}$, giving the coefficients in terms of p and q .	4
(ii)	Deduce that if $\gamma = \alpha\beta$ then $(3p - q)^2 + q = 0$	2
(c)	The polynomial $P(x)$ leaves a remainder of 9 when divided by $(x - 2)$ and a remainder of 4 when divided by $(x - 3)$. Find the remainder when $P(x)$ is divided by $(x - 2)(x - 3)$.	3
(d)	(i) Solve $Z^5 = 1$ over the complex field giving your answers in modulus-argument form.	2
(ii)	Hence write $Z^5 - 1$ as the product of linear and quadratic factors.	2

End of Question 14

Question 15 (15 marks) Begin a NEW sheet of paper.**Marks**

- (a) A mass of 1kg is falling under gravity (g) through a medium in which the resistance to the motion is proportional to the square of the velocity. (k is the constant of proportionality)

(i) Draw a sketch showing all forces acting. 1

(ii) Write an equation for the acceleration of this mass. 1

(iii) Show that the mass reaches a terminal velocity 1

$$\text{given by } v = \sqrt{\frac{g}{k}}.$$

(iv) Show that the distance it has fallen when it reaches 4

$$\text{a velocity } v \text{ m/s is given by } x = \frac{1}{2k} \ln \left(\frac{g}{g - kv^2} \right)$$

- (b) (i) Show that the recurrence (reduction) formula for

$$I_n = \int \sec^n x dx \quad \text{4}$$

$$\text{is } I_n = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2}$$

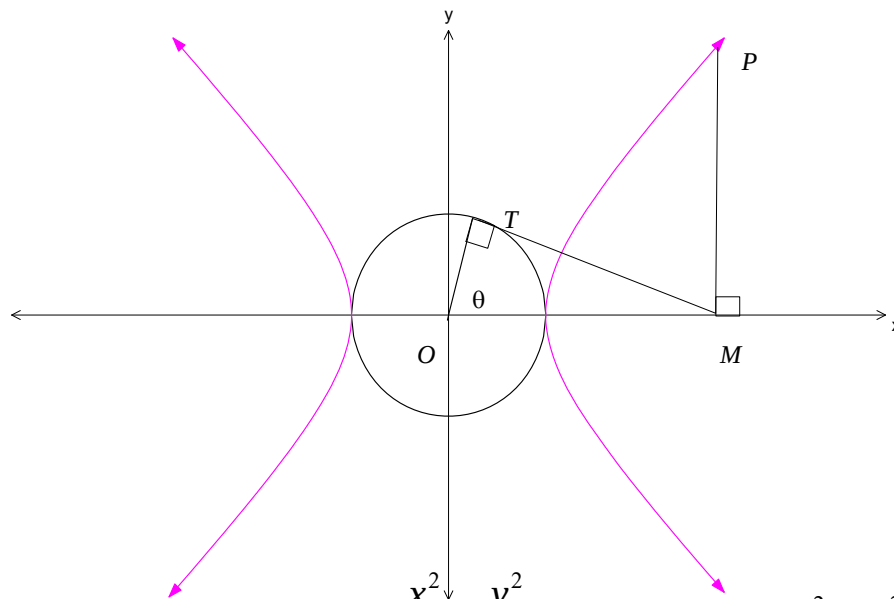
(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x dx$ 2

- (c) A cubic equation in z has all real coefficients. If two of the roots are 3 and $2 + i$ determine the equation. 2

End of Question 15

Question 16 (15 marks) Begin a NEW sheet of paper.**Marks**

(a)



The sketch shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = a^2$

with $a, b \geq 0$. T lies on the circle where $\angle TOX = \theta$ and $0 \leq \theta \leq \frac{\pi}{2}$. The tangent at T meets OX at M and MP is perpendicular to OX with P on the hyperbola. Coordinates of T are $(a \cos \theta, a \sin \theta)$

(i) Find the equation of the tangent TM and hence the coordinates of M. 3

(ii) Hence show that the coordinates of P are $(a \sec \theta, b \tan \theta)$ 1

(iii) If $Q(a \sec \beta, b \tan \beta)$ is another point on the hyperbola, where $\theta + \beta = \frac{\pi}{2}$ and $\theta \neq \frac{\pi}{4}$, show that the equation of PQ is $ay = b(\cos \theta + \sin \theta)x - ab$. 3

(iv) Every such chord PQ passes through a fixed point, find its coordinates. 2

(v) Show that as θ approaches $\frac{\pi}{2}$, PQ approaches a line parallel to an asymptote. 2

Question 7 continues on page 9.

Question 16 continued.**Marks**

- (b) (i) By letting $Z = \cos \theta + i \sin \theta$ show that

1

$$Z^n + \frac{1}{Z_n} = 2 \cos n\theta .$$

- (ii) Hence express $\cos^4 \theta$ in terms of $\cos n\theta$

3**End of Question 16****End of Examination**

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

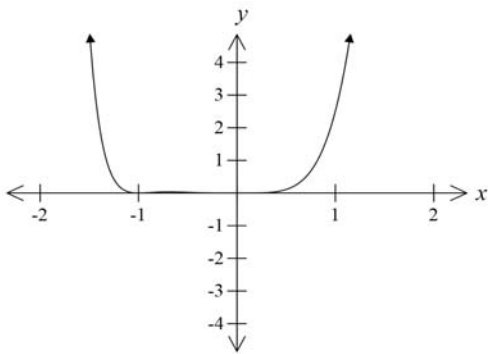
NOTE : $\ln x = \log_e x, \quad x > 0$

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**2015
TRIAL HSC
EXAMINATION**

Mathematics
Extension 2

SOLUTIONS

Section I	Trial HSC Examination- Mathematics Extension 2	2015
1		1 Mark: D
2	$b^2 = a^2(e^2 - 1) \qquad a^2 = 144 \text{ and } b^2 = 25.$ $25 = 144(e^2 - 1) \qquad a = 12 \qquad b = 5$ $(e^2 - 1) = \frac{25}{144} \text{ or } e^2 = \frac{169}{144} \text{ or } e = \frac{13}{12}$ Equation of the directrices are $x = \pm \frac{a}{e} = \pm \frac{144}{13}.$	1 Mark: D
3	$z - i \leq 2$ represents a region with a centre is (0, 1) and radius is less than or equal to 2. $0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$ represents a region between angle 0 and $\frac{3\pi}{4}$ whose vertex is (1, 0), not including the vertex $z - i \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$	1 Mark: A
4	$\int \frac{2}{x^2 + 4x + 13} dx = 2 \int \frac{dx}{(x + 2)^2 + 3^2}$ $= \frac{2}{3} \tan^{-1} \frac{(x + 2)}{3} + c$	1 Mark: B

5	To find the equation of tangent through P $x = a \cos \theta \qquad y = b \sin \theta$ $\frac{dx}{d\theta} = -a \sin \theta \qquad \frac{dy}{d\theta} = b \cos \theta$ $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$ $= b \cos \theta \times \frac{1}{-a \sin \theta} = \frac{-b \cos \theta}{a \sin \theta}$ Equation of the tangent $y - y_1 = m(x - x_1)$ $y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$ $ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$ $bx \cos \theta + ay \sin \theta = ab(\sin^2 \theta + \cos^2 \theta)$ $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$	1 Mark: D
6	The base is an annulus. $A = \pi(r_2^2 - r_1^2)$ $= \pi(r_2 + r_1)(r_2 - r_1)$ $(r-1)^2 + \frac{y^2}{4} = 1$ $r^2 - 2r + \frac{y^2}{4} = 0$ $r = \frac{2 \pm \sqrt{(-2)^2 - 4 \times 1 \times \frac{y^2}{4}}}{2}$ $r = 1 \pm \sqrt{1 - \frac{y^2}{4}}$ Therefore $r_2 + r_1 = 2$ and $r_2 - r_1 = 2\sqrt{1 - \frac{y^2}{4}}$ $V = \lim_{\delta y \rightarrow 0} \sum_{y=-2}^2 \pi \times 2 \times 2\sqrt{1 - \frac{y^2}{4}} \delta y$ $= \int_{-2}^2 2\pi \sqrt{4 - y^2} dy$	1 Mark: D

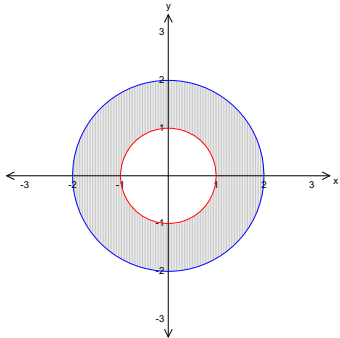
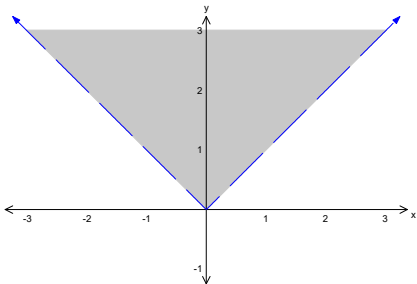
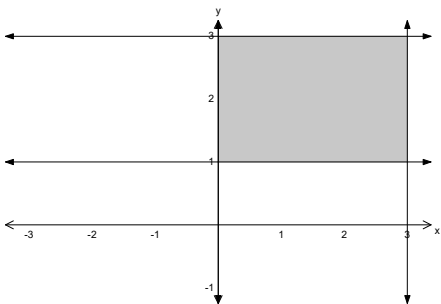
7	$x = \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ $\alpha = \frac{1}{x} \text{ satisfies } x^3 - 3x^2 - x + 2 = 0$ $\left(\frac{1}{x}\right)^3 - 3\left(\frac{1}{x}\right)^2 - \frac{1}{x} + 2 = 0$ $1 - 3x - x^2 + 2x^3 = 0$ $2x^3 - x^2 - 3x + 1 = 0$	1 Mark: C
8	$y = \sin^{-1} x - \sqrt{1-x^2}$ $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \times -2x$ $= \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}$ $= \frac{1+x}{\sqrt{1-x^2}}$ $= \frac{1+x}{\sqrt{(1+x)(1-x)}}$ $= \frac{\sqrt{1+x}}{\sqrt{1-x}}$ Result defined for $-1 \leq x \leq 1$	1 Mark: A
9	Using the conjugate root theorem $1+i$ and $1-i$ are both roots of the equation $z^3 + pz + q = 0$. $(1+i) + (1-i) + \alpha = 0 \quad (\text{sum of the roots})$ $\alpha = -2$ $(1+i) \times (1-i) \times -2 = -q \quad (\text{product of the roots})$ $(1+1) \times -2 = -q$ $q = 4$ $(1+i)(1-i) + (1-i) - 2 + (1+i) - 2 = p$ $p = -2$ Therefore $p = -2$ and $q = 4$	1 Mark: B

10	$F = \frac{m}{x^3}(6-10x)$ $ma = \frac{m}{x^3}(6-10x)$ $v \frac{dv}{dx} = \frac{6}{x^3} - \frac{10}{x^2}$ $\int v dv = \int \left(\frac{6}{x^3} - \frac{10}{x^2} \right) dx$ $\frac{1}{2} v^2 = \left(\frac{6x^{-2}}{-2} - \frac{10x^{-1}}{-1} \right) + c$ $\frac{1}{2} v^2 = \left(\frac{-3}{x^2} + \frac{10}{x} \right) + c$ When $v = 0$ and $x = 1$ $\frac{1}{2} 0^2 = \left(\frac{-3}{1^2} + \frac{10}{1} \right) + c$ $c = -7$ $\frac{1}{2} v^2 = \left(\frac{-3}{x^2} + \frac{10}{x} \right) - 7$ $v^2 = \left(\frac{-6}{x^2} + \frac{20}{x} \right) - 14$ $= \frac{-6 + 20x - 14x^2}{x^2}$ $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$	1 Mark: C
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Question 11		Trial HSC Examination- Mathematics Extension 2		2015	
Part	Solution			Marks	Comment
(a)	Let $\frac{A}{x+1} + \frac{B}{x-1} = \frac{1}{x^2-1}$			1	Any method to find A and B
	$\therefore A(x-1) + B(x+1) = 1$				
	If $x = 1$ $2B = 1$				
	$B = \frac{1}{2}$				
	If $x = -1$ $2A = -1$				
	$A = -\frac{1}{2}$			1	
	$\frac{dx}{x^2-1} = \frac{1}{2} \int \frac{1}{x-1} - \frac{1}{x+1} dx$			1	
	$\therefore \int \quad = \frac{1}{2} [\ln(x-1) - \ln(x+1)] + C$				No mark for C
	$= \frac{1}{2} \ln\left(\frac{x-1}{x+1}\right) + C$			1	
	$= \ln \sqrt{\frac{x-1}{x+1}} + c$				Total = 4

Question 11		Trial HSC Examination- Mathematics Extension 2		2015	
Part	Solution	Marks	Comment		
(b)	When $x = 1 \quad \theta = 0$ $x = 0 \quad \theta = \frac{\pi}{2}$ Let $x = \cos \theta$ $dx = -\sin \theta d\theta$ $\therefore \int_0^1 \sqrt{1-x^2} dx = \int_{\frac{\pi}{2}}^0 \sqrt{1-\cos^2 \theta} (-\sin \theta d\theta)$ $= \int_{\frac{\pi}{2}}^0 -\sin^2 \theta d\theta$ $= \frac{1}{2} \int_{\frac{\pi}{2}}^0 (\cos 2\theta - 1) d\theta$ $= \frac{1}{2} \left[\frac{1}{2} \sin 2\theta - \theta \right]_{\frac{\pi}{2}}^0$ $= \frac{1}{2} \left[(0) - \left(0 - \frac{\pi}{2} \right) \right]$ $= \frac{\pi}{4}$	1 1 1 1	Can use alternate methods ie $x = \sin \theta$ Total = 4		
(c)	$I = \int e^x \sin x dx$ $= e^x \sin x - \int e^x \cos x dx$ $= e^x \sin x - (e^x \cos x - \int e^x [-\sin x] dx)$ $= e^x \sin x - e^x \cos x - I$ $2I = e^x (\sin x - \cos x) + C$ $\therefore I = \frac{e^x}{2} (\sin x - \cos x) + C$	1 1 1 1	Total = 4		

Question 11		Trial HSC Examination- Mathematics Extension 2		2015	
Part	Solution			Marks	Comment
(d)	$\int_0^{\frac{\pi}{2}} \cos x \sin^3 x \, dx = \left[\frac{1}{4} \sin^4 x \right]_0^{\frac{\pi}{2}}$			1	May be done by a substitution of $u = \sin x$ $du = \cos x \, dx$
	$= \frac{1}{4} \left[\sin^4 \left(\frac{\pi}{2} \right) - \sin^4 (0) \right]$			1	
	$= \frac{1}{4} [1 - 0]$			1	Total = 3
	$= \frac{1}{4}$				

Question 12		Trial HSC Examination- Mathematics Extension 2	2015
Part	Solution	Marks	Comment
(v)	$B^4 = 2^4 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$ $= 16 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$ $= -8 + 8\sqrt{3}i$	1	
(b) (i)		1	
(ii)		1	Dotted lines
(iii)		2	
(c) (i)	$OA = 2(-y + ix)$	1	
(ii)	$OB = OA + AB$ $= -2y + 2ix + x + iy$ $= (x - 2y) + (2x + y)i$	1	
(iii)	$BC = -OA$ $= 2y - 2xi$	1	

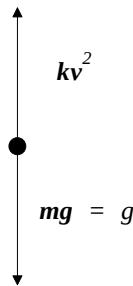
Question 13		Trial HSC Examination- Mathematics Extension 2		2015
Part	Solution	Marks	Comment	
(a)	$x^2 + 3xy - y^2 = 13$ $2x + 3y + 3x \cdot \frac{dy}{dx} - 2y \cdot \frac{dy}{dx} = 0$ $\frac{dy}{dx}(3x - 2y) = -(2x + 3y)$ $\frac{dy}{dx} = \frac{-(2x + 3y)}{(3x - 2y)}$ $\therefore \text{at } (2,3) \quad \frac{dy}{dx} = -\frac{(4 + 9)}{6 - 6}$ $\therefore \text{Gradient infinite}$ $\therefore \text{Tangent is vertical}$	2 2	Total = 4	

(b)			
(i)	$a = b = 2y$ and $\theta = 60^\circ$ $Area = \frac{1}{2} a b \sin C$ $Area = \frac{1}{2} \times 2y \times 2y \times \sin 60^\circ$ $Area = \frac{1}{2} \times 4y^2 \times \frac{\sqrt{3}}{2}$ $Area = \sqrt{3} y^2$ $Area = \sqrt{3} (16 - x^2)$		

(c) (i)	In $HPRM$ $\angle HPR = \angle RMH = 90^\circ$ (Angle between radius and tangent) $\therefore HPRM$ is a cyclic quadrilateral (Opposite angles supplementary) Similarly for $MRQK$	1 1	Total = 2
(ii)	Join PM and QM. in $\triangle PRM$ and $\triangle MQK$ $\angle PRM = \angle MKQ$ (Exterior angle of cyclic quad $MRQK$) $PR = RM$ (Tangent from external point) $KM = KQ$ (radii) $\therefore$ Triangle isosceles. $\therefore \angle RPM = \angle RMP = \angle KMQ$ $\therefore \triangle PRM \cong \triangle MQK$ (Equiangular)	1 1	Total = 2

Question 14		Trial HSC Examination- Mathematics Extension 2	2015	
Part	Solution	Marks	Comment	
(b) (ii)	For $\gamma = \alpha\beta$ $\frac{\alpha\beta}{\gamma} = 1$ is a root $\therefore 1 + \frac{9p^2}{q} - 6p + q = 0$ $q + 9p^2 - 6pq + q^2 = 0$ $\therefore (3p - q)^2 + q = 0$	1 1	Total = 2	
(c)	$P(x) = (x - 2)(x - 3)Q(x) + (ax + B)$ $P(2) = 9$ $P(3) = 4$ Therefore $2a + b = 9$ $3a + b = 4$ $a = -5, b = 19$	1 1 1	Total = 3	
(d)	Let $Z = cis\theta$ $Z^5 = cis5\theta$ $\therefore \cos 5\theta + i \sin 5\theta = 0$ $\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$ i.e. $= 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{-4\pi}{5}, \frac{-2\pi}{5}$ $\therefore$ Roots are $Z_1 = 1, \quad Z_2 = cis \frac{2\pi}{5}, \quad Z_3 = cis \frac{4\pi}{5}$ $Z_4 = cis(-\frac{4\pi}{5}) \quad Z_5 = cis(-\frac{2\pi}{5})$ $\quad \quad \quad = \bar{Z}_3 \quad \quad \quad = \bar{Z}_2$	1 1	Total = 2	

Question 14		Trial HSC Examination- Mathematics Extension 2		2015	
Part	Solution			Marks	Comment
(e) ii)	$z^5 - 1$				
	$= (z - z_1)(z - z_2)(z - z_5)(z - z_3)(z - z_4)$				
	$= (z - z_1)(z^2 - (z_2 + z_2z_5)z + z_2z_5)(z^2 - (z_3 + z_4)z + z_3z_4)$			1	
	$= (z - 1)(z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1)$			1	
					Total = 2

Question 15		Trial HSC Examination- Mathematics Extension 2		2015	
Part	Solution	Marks	Comment		
(a) (i)		1			
(a) (ii)	$\ddot{x} = g - kv^2$	1			
(a) (iii)	When $\ddot{x} = 0$ $g = kv^2$ $\therefore v = \sqrt{\frac{g}{k}}$	1			
(a) (iv)	$\ddot{x} \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = g - kv^2$ $= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \cdot \frac{dv}{dx} = g - kv^2$ $= v \cdot \frac{dv}{dx} = g - kv^2$ $\therefore \frac{dv}{dx} = \frac{g - kv^2}{v}$ $\frac{dx}{dv} = \frac{v}{g - kv^2}$ $\therefore x = -\frac{1}{2k} \ln(g - kv^2) + c$ When $x = 0$ $v = 0$ $\therefore c = \frac{1}{2k} \ln g$ $\therefore x = -\frac{1}{2k} \ln(g - kv^2) + \frac{1}{2k} \ln g$ $= \frac{1}{2k} \ln \left(\frac{g}{g - kv^2} \right)$	1 <			

Question 15		Trial HSC Examination- Mathematics Extension 2	2015	
Part	Solution	Marks	Comment	
(b) (i)	$I_n = \int \sec^n x dx$ $= \int \sec^{n-2} x \cdot \sec^2 x dx$ $= \tan x \cdot \sec^{n-2} x - \int (n-2) \sec^{n-3} x \cdot \tan x \cdot \sec x \tan x dx$ $= \tan x \sec^{n-2} x - (n-2) \int \sec^{n-2} x \cdot \tan^2 x dx$ $= \tan x \sec^{n-2} x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) dx$ $= \tan x \sec^{n-2} x - (n-2) \int (\sec^n x - \sec^{n-2} x) dx$ $= \tan x \sec^{n-2} x - (n-2) I_n + (n-2) I_{n-2}$ $I_n + (n-2) I_n = (n-1) I_n = \tan x \sec^{n-2} x + (n-2) I_{n-2}$ $I_n = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2}$	1 1 1 1		
		1	Total = 4	
(b) (ii)	$\int_0^{\frac{\pi}{4}} \sec^4 x dx = I_4 = \left[\frac{1}{3} \tan x \sec^2 x + \frac{2}{3} \int \sec^2 x \right]_0^{\frac{\pi}{4}}$ $= \left[\frac{1}{3} \tan x \sec^2 x + \frac{2}{3} \tan x \right]_0^{\frac{\pi}{4}}$ $= \left(\frac{1}{3} \tan \frac{\pi}{4} \sec^2 \frac{\pi}{4} + \frac{2}{3} \tan \frac{\pi}{4} \right) - \left(\frac{1}{3} \tan 0 \sec^2 0 + \frac{2}{3} \tan 0 \right)$ $= \left(\frac{1}{3} \times 1 \times 2 + \frac{2}{3} \times 1 \right) - 0$ $= \frac{4}{3}$	1 1		
		1	Total = 2	

Question 15		Trial HSC Examination- Mathematics Extension 2		2015	
Part	Solution	Marks		Comment	
(c)	If one root is $2 + i$, another is $2 - i$ $\therefore (z - (2 + i))(z - (2 - i))(z - 3) = 0$ $(z - 2 - i)(z - 2 + i)(z - 3) = 0$ $(z^2 - 4z + 5)(z - 3) = 0$ $z^3 - 7z^2 + 17z - 15 = 0$	1			
		1		Total = 2	

Question 16		Trial HSC Examination- Mathematics Extension 2		2015	
Part	Solution	Marks	Comment		
(a) (i)	Coordinates of T are $(a \cos \theta, a \sin \theta)$ $x^2 + y^2 = a^2$ $\therefore 2x + 2y \frac{dy}{dx} = 0$ $\therefore \frac{dy}{dx} = -\frac{x}{y}$ at T $\frac{dy}{dx} = -\frac{\cos \theta}{\sin \theta}$ $\therefore$ Equation TM $y - a \sin \theta = -\frac{\cos \theta}{\sin \theta}(x - a \cos \theta)$ $x \cos \theta + y \sin \theta = a$ When $y = 0$ $x = a \sec \theta$ $\therefore$ Coordinates M are $(a \sec \theta, 0)$	1 			

(a) iii)	$a \sec \beta = a \sec\left(\frac{\pi}{2} - \theta\right)$ $= a \operatorname{cosec} \theta$ $b \tan \beta = b \tan\left(\frac{\pi}{2} - \theta\right)$ $= b \cot \theta$ $\text{Gradient } PQ = \frac{b \cot \theta - b \tan \theta}{a \operatorname{cosec} \theta - a \sec \theta}$ $= \frac{b}{a} \left\{ \frac{\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}}{\frac{1}{\sin \theta} - \frac{1}{\cos \theta}} \right\}$ $= \frac{b}{a} \left\{ \frac{\frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}}{\frac{\cos \theta - \sin \theta}{\sin \theta \cos \theta}} \right\}$ $= \frac{b}{a} \left(\frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta - \sin \theta} \right)$ $= \frac{b}{a} (\cos \theta + \sin \theta)$ $\therefore$ Equation PQ is $y - b \tan \theta = \frac{b}{a} (\cos \theta + \sin \theta) (x - a \sec \theta)$ $y - b \tan \theta = \frac{b}{a} (\cos \theta + \sin \theta) x - b \cos \theta \sec \theta - b \sin \theta \sec \theta$ $y - b \tan \theta = \frac{b}{a} (\cos \theta + \sin \theta) x - b - b \tan \theta$ $y = \frac{b}{a} (\cos \theta + \sin \theta) x - b$ $ay = b (\cos \theta + \sin \theta) x - ab$	1 1 1	Total = 3
(a) (iv)	All of the lines have the same y intercept. i.e. $y = -b$ $\therefore$ The fixed point is the intercept $(0, -b)$	1 1	Total = 2

(a) (v)	Equations of Asymptotes are $y = \pm \frac{b}{a} x$ Gradients of asymptotes are $m = \pm \frac{b}{a}$ As $\theta \rightarrow \frac{\pi}{2}$ the equation of $PQ \rightarrow ay = b(0+1)x - ab$ $\therefore$ Gradient of $PQ \rightarrow \frac{b}{a}$ $\therefore PQ$ approaches a line which is parallel to an asymptote.	1	
(b) i)	$z = \cos \theta + i \sin \theta$ $\frac{1}{z} = z^{-1} = \cos \theta - i \sin \theta$ By De Moivres Theorem $z^n = \cos n\theta + i \sin n\theta$ $z^{-n} = \cos n\theta - i \sin n\theta$ $z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$ $z^n + \frac{1}{z^n} = 2 \cos n\theta$	1	
ii)	$\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^3 \frac{1}{z} + 6z^2 \frac{1}{z^2} + 4z \frac{1}{z^3} + \frac{1}{z^4}$ $= \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6$ $(2 \cos \theta)^4 = 2 \cos 4\theta + 4(2 \cos 2\theta) + 6$ $2^4 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$ $\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$	1	
		1	
		1	Total = 3